Traffic Simulation of Kobe-city

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Abstract

A traffic simulation is carried out as the first step for conducting a virtual social experiment of Kobe-city. In order to simulate an actual traffic flow, a road structure was built utilizing a high-quality digital map data, and an origin-destination information of vehicles was determined by a geographical population distribution. However, the traffic flow obtained from the simulation was deviated from that of the Kobe traffic census data. In the presentation, we are going to present our attempts to resolve these difficulties.

Keyword: Social simulation, Traffic flow, Network, Big-data

1. Background

The trial site of Kobe in the Center of Innovation(COI STREAM: Science and Technology based Radical Innovation and Entrepreneurship Program of the Ministry of Education, Culture, Sports, Science and Technology of Japan) is working on developing and providing decision support system which utilizing K computer to realize a new public health system. Our team focuses on the infection disease. To develop a technology which predicts infection disease, we are trying to conduct a virtual social experiment by a computer simulation.

To conduct the virtual social experiment about infection disease, it is necessary to take into account a traffic flow as well as a flow of people. This is because cars and trains play major roles in today's urban mobility. Therefore, in this study, we tried to perform a traffic simulation of Kobe-city as a first step of the virtual social experiment.

2. Method

Although various traffic simulators exist, SUMO (Simulation of Urban Mobility) [1] was adopted because it is fast enough to conduct a large-scale simulation in a practical time. A method of the traffic simulation utilizing SUMO consists of building a road structure and determining an origin-destination (OD) information of vehicles. In this study, the former was constructed from a high-quality digital map data provided from Zenrin Co. Ltd., and the latter was determined by a geographical population distribution estimated by NTT DoCoMo, Inc. from its mobile phone call data. Since the OD information is not uniquely determined from the population distribution, we are trying to adjust free parameters so that the simulations are compatible with the traffic census data taken by the Ministry of Land, Infrastructure, Transport and Tourism of Japan. This adjustment is challenging because there are many free parameters and the response to the parameter change is nonlinear.

3. Results and Discussions

As described in the previous section, vehicles were generated in proportion to the population distribution. However, the amount of traffic flows obtained from the simulation were deviated from that of the traffic census data. Moreover, artificial gridlocks occurred in the simulation due to unrealistic routing.

In order to resolve these problems, routing results were investigated. The results indicate that the gridlocks occur at the places that many vehicles passing at the same time. Therefore, it is necessary to calculate the appropriate routes which are taken into account a weight calculated from a prediction of gridlocks utilizing routing results.

4. Summary

To conduct the virtual social experiments, a traffic simulation was performed using SUMO. The simulations are carried out utilizing the high-quality digital map data and the geographical population distribution. In order to simulate the actual traffic flow, the problems mentioned above have to be resolved.

Acknowledgement

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References

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